

Response-Aware Marginal-Utility Routing for Multi-Context Prediction

Xinling Wen^a, Jiahao Zhang^{a,*}, Yu Chen^{a,**}

*^aSchool of Electronics and Information, Zhengzhou University of
Aeronautics, Zhengzhou, China*

Abstract

Auxiliary contexts can improve predictive models, yet the value of a context changes as other contexts are selected. We formulate multi-context selection through state-conditioned marginal predictive utility, defined by the change in end-task loss produced by adding a candidate to the current context set. This formulation exposes decision value that cannot be represented by a fixed standalone score. We introduce Response-Aware Marginal-Utility Routing (R-MUR), which first screens candidates from cached representations and then evaluates predictor responses only for a small retained set. Compact response summaries are used to estimate marginal utility, while a regression and ranking objective aligns utility estimation with sequential selection. A decision analysis connects estimation error to routing regret. Controlled studies show how redundancy and candidate-set growth change the selection problem, and evaluations on standard traffic benchmarks show the method significantly ahead of the strongest of fourteen competitors on three of six benchmarks, unresolved on two and below simple pooling on one, and ahead of every similarity-, diversity-, coverage- and utility-based selection policy on all six. The results show that state-conditioned marginal utility provides a

*Corresponding author
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Email addresses: wenxinling@zua.edu.cn (Xinling Wen),
marchematics@gmail.com (Jiahao Zhang), chenyu@zua.edu.cn (Yu Chen)

useful decision principle for budgeted multi-context prediction.

5 *Keywords:* context selection, marginal utility, context routing, negative
6 transfer, predictive decision support

7 1. Introduction

8 Predictive systems increasingly combine a primary input with auxiliary
9 contexts, such as retrieved examples, historical observations, or records from
10 neighbouring sensors. The usefulness of a context depends on what the pre-
11 dictor has already received. A context that improves a prediction on its own
12 can become redundant once similar information has been selected, while a
13 context with a modest standalone score can supply exactly the information
14 that is still missing. The value of a candidate is therefore a property of the
15 selected set, not of the candidate alone.

Existing selection rules derive candidate value from different signals. Relevance measures compatibility with the query, and diversity or coverage methods update a content-based objective as the selected set grows. Standalone predictive utility instead measures the effect of a candidate on the predictor relative to an empty context set. None of these quantities directly represents the change in end-task loss produced by adding a candidate to the contexts already selected. We define that quantity, the marginal predictive utility of a candidate, as the reduction in end-task loss obtained by adding it to the context set selected so far,

$$m(j \mid A) = F(A \cup \{j\}) - F(A), \quad (1)$$

16 where A is the current set and F measures the expected loss reduction relative
17 to a predictor that receives no auxiliary context. A fixed score does not

18 determine this quantity, because adding a candidate changes the prediction
19 that the next candidate acts on.

20 A perfect standalone ranking fixes the order of candidates in advance,
21 while a sequential rule re-evaluates candidate value after every addition. The
22 two policies can disagree even when the standalone scores are exact, and
23 the loss reduction that only the sequential rule obtains measures how much
24 decision value the fixed-score view leaves unrepresented. We call this quantity
25 structural headroom and treat it as a property of the prediction task rather
26 than of a particular router.

27 Two properties make marginal utility difficult to learn. First, as the
28 candidate pool grows, the utility gap between the best alternatives shrinks,
29 so a model with stable average error becomes less able to identify the best
30 candidate. Second, cached context representations describe the content of
31 a candidate, not its effect on the predictor: two candidates with similar
32 representations can change the prediction in different directions, and only
33 an evaluation of the predictor reveals which one reduces the loss. These
34 properties set the design requirements for a practical router.

35 We introduce Response-Aware Marginal-Utility Routing (R-MUR). A
36 cached score first narrows the candidate pool to a small shortlist. The fixed
37 predictor is then evaluated on the shortlisted candidates only, and compact
38 summaries of its response are combined with cached representations to esti-
39 mate marginal utility. Training combines regression on observed marginals
40 with a pairwise ranking term weighted by the utility gap, which aligns esti-
41 mation with the selection decision. The design follows directly from the two
42 properties above: the response summaries expose the prediction change that

enters the marginal, and the pairwise objective aligns utility estimation with the candidate ordering that selection actually uses.

The paper contributes a decision formulation of multi-context prediction in which candidate value is defined by its marginal effect on a fixed predictor, and it derives the loss decomposition and the two oracle policies that make state conditioning measurable. It contributes a routing method that learns this quantity from a small number of predictor evaluations per decision, and a decision analysis that relates estimation error to one-step routing regret. Finally, it contributes a controlled study of when state conditioning matters, together with a multi-target evaluation on standard traffic benchmarks in which the method is compared against similarity-, diversity-, coverage-, determinantal, and utility-based selection on the same candidate pools and prediction targets, including the trivial policy of using every candidate, and in which it achieves the best average rank while being significantly ahead of the strongest competitor on three benchmarks, unresolved on two and below on one.

Scope and companion manuscripts. This paper is the method paper: it defines the decision object, develops the estimator, and evaluates it against competing selection rules under a fixed protocol. Two companion manuscripts extend it in orthogonal directions and are cited here only where their results bear on the method’s scope: one develops the identifiability theory of marginal utility — when the quantity can be estimated from observable response, and why stronger fixed predictors reduce the value that routing can realise — and one reports the boundary evidence, namely the behaviour under a subset-capable nonlinear backbone, a second task family, cross-dataset

68 generalisation and the compute profile. The protocol, the predictors and the
69 baselines are shared across the three papers by design, so that their claims are
70 directly comparable; no result reported here depends on either companion
71 for its support.

72 **2. Related Work**

73 Selection of auxiliary contexts has been studied mainly through the con-
74 tent of the candidates. Retrieval and demonstration selection choose exam-
75 ples for a frozen predictor [1, 2], and many selection rules score candidates
76 by similarity to the query [3–6]. Other work studies prompt-order sensitivity
77 and calibration [7, 8], retrieval-augmented prediction [9–15], and the effec-
78 t of context composition on how a model uses retrieved information [16].
79 Recent KBS studies likewise make external information acquisition adap-
80 tive, including structured engineering-knowledge retrieval [17], causal and
81 conflict-aware retrieval [18], adaptive multi-hop evidence retrieval [19], and
82 pipelines that adapt the granularity of the retrieved context [20]. A second
83 family enforces diversity or coverage, using penalties for repeated content or
84 submodular objectives [21–23], and listwise rerankers compare candidate sets
85 directly [24–27]. These methods determine which candidates are relevant or
86 complementary in content. Relevance is computed from the candidate and
87 the query alone, while diversity and coverage objectives are set functions of
88 the selected content; none of them is defined through the effect of a candidate
89 on the predictor’s loss.

90 A smaller line of work values candidates through their effect on the predic-
91 tion. Outcome-aware selection estimates the incremental utility of a demon-

92 stration relative to an empty prompt [28], and data valuation and influence
 93 estimation attribute a model’s output to training examples [29–32]. These
 94 approaches define value with respect to the empty set or to a fixed reference,
 95 which is the standalone utility $u(j \mid \emptyset)$ in our notation. The difficulty of
 96 using auxiliary information well is also visible in recent KBS work on nega-
 97 tive transfer, where task-specific information is relearned to suppress harmful
 98 cross-task transfer [33] and source-domain knowledge is selectively integrated
 99 before distillation [34]. Such methods decide what should be transferred at
 100 the task or domain level; our object is the candidate-level marginal $m(j \mid A)$
 101 with respect to a nonempty, evolving selected set, and it coincides with the
 102 standalone utility only at the first selection.

103 Set-selection methods supply the structural counterpart. Submodular
 104 objectives formalize coverage, diversity, and diminishing returns [35–37], de-
 105 terminantal point processes model diverse subsets probabilistically [38], and
 106 adaptive submodularity and influence maximization extend the tools to se-
 107 quential decisions [39, 40]. KBS has also studied adaptive subset construction
 108 through feature-selection objectives, including model-independent diversity-
 109 based importance [41] and graph-regularized selection under partial super-
 110 vision [42]; such objectives select informative or nonredundant input dimen-
 111 sions, whereas R-MUR evaluates an auxiliary context through its incremental
 112 effect on the downstream predictor. Classical monotone formulations reward
 113 every additional element, and a context can instead increase the loss, so the
 114 marginal utility we estimate may be negative; this is the phenomenon that
 115 the transfer literature calls negative transfer [43–45].

116 Learned routing treats selection as a decision problem. Mixture-of-

117 experts systems route inputs to expert subsets [46–48], sparse and soft
 118 routing architectures reduce computation while increasing capacity [49–52],
 119 and routing networks select computation paths across tasks [53, 54]. Adap-
 120 tive routing has recently appeared in KBS across several settings: expert
 121 activation conditioned on temporal behaviour patterns [55], expert routing
 122 by attention over multimodal state [56], sparse expert selection for collabora-
 123 tive perception driven by agent-specific information [57], in-context experts
 124 for sequential recommendation [58], and query-specific intervention selec-
 125 tion from model-behaviour signals [59]. These studies establish the value of
 126 adaptive routing, but the routed object is an expert, an intervention, or a
 127 computational component. Decision-focused learning links prediction quality
 128 to downstream decision quality [60–63], and sequential selection is related to
 129 bandit exploration and off-policy value estimation [64, 65]. R-MUR belongs
 130 to this line but routes auxiliary contexts rather than computation or model
 131 components, and it keeps the predictor fixed so that the routing target is
 132 their state-conditioned marginal effect on end-task loss.

133 3. State-Conditioned Context Selection

134 Let S_0 be an anchor context, x a query, y its target, and $\mathcal{C} = \{C_1, \dots, C_K\}$
 135 a candidate pool. A selected set is $A \subseteq \{1, \dots, K\}$. A predictor produces
 136 $f_A(x)$ from S_0 and the contexts indexed by A ; the predictor is trained before
 137 routing and kept fixed during context selection and router training.

We define the net set utility as

$$F(A) = \mathbb{E}[\ell(f_\emptyset(x), y) - \ell(f_A(x), y)] - \lambda c(A), \quad (2)$$

where $c(A)$ is a context cost. The marginal utility of a candidate is

$$m(j \mid A) = F(A \cup \{j\}) - F(A). \quad (3)$$

For one observed episode, the training target is the counterfactual loss difference

$$\tilde{m}(j \mid A) = \ell(f_A(x), y) - \ell(f_{A \cup \{j\}}(x), y) - \lambda c_j. \quad (4)$$

138 This is a realized observation of the marginal. Its interpretation as an ex-
139 pectation requires a fixed predictor and a specified state-sampling rule.

For squared loss, let $\mu = \mathbb{E}[y \mid S_0, x, \mathcal{C}]$ and let $d_{A,j} = f_{A \cup \{j\}}(x) - f_A(x)$. The marginal separates into

$$m(j \mid A) = 2d_{A,j}(\mu - f_A(x)) - d_{A,j}^2 - \lambda c_j. \quad (5)$$

140 The first term is the interaction between the current residual and the direc-
141 tion of the candidate-induced correction, the second term penalizes overcor-
142 rection, and the third accounts for cost. Relevance and standalone candidate
143 scores do not determine these three terms, and the same candidate can be
144 useful at one state and harmful at another.

Two oracle policies expose the structural value of this state dependence. The standalone oracle ranks candidates by the true standalone marginal $m(j \mid \emptyset)$ and keeps that order fixed. The sequential oracle recomputes the true marginal after every addition. Let G_{stand} and G_{seq} denote their prediction gains over the anchor-only predictor. We measure structural headroom as

$$H_{\text{state}} = \frac{G_{\text{seq}} - G_{\text{stand}}}{G_{\text{seq}}}. \quad (6)$$

145 This quantity is an evaluation diagnostic. It isolates the value created by
146 conditioning on the selected set, independently of router estimation error. It

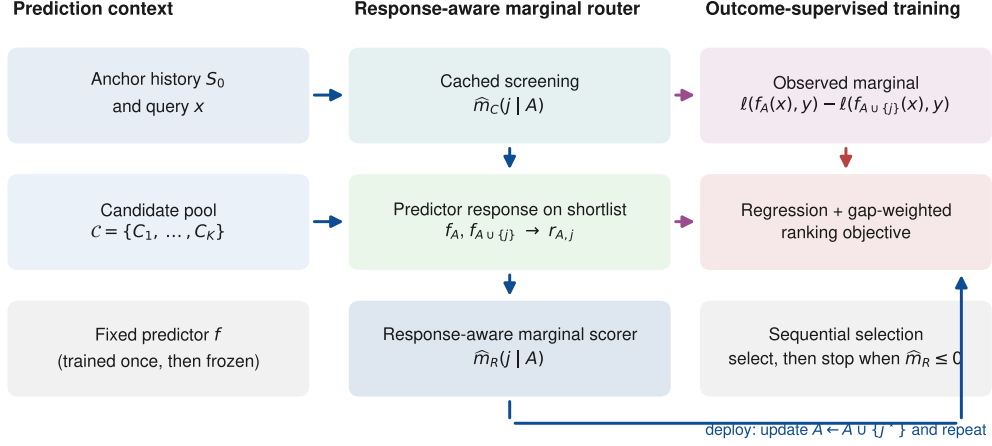


Figure 1: Response-aware marginal-utility routing. The router screens the candidate pool with cached representations, evaluates the frozen predictor on a shortlist, and estimates marginal utility from the predictor response. Training uses the observed marginal utility of each candidate as supervision, and deployment selects sequentially, stopping when no candidate is estimated to be useful.

147 is defined when the sequential oracle improves on the anchor, which holds
 148 in every evaluation we report; the unnormalised difference $G_{\text{seq}} - G_{\text{stand}}$ is
 149 reported alongside the ratio so that the normalised value can be read together
 150 with the scale of the opportunity it normalises.

151 4. Response-Aware Marginal-Utility Routing

152 R-MUR estimates the marginal utility of a candidate with a fixed predic-
 153 tor. It screens candidates with cached representations, evaluates the predic-
 154 tor only on a shortlist, and estimates the marginal from the cached context
 155 together with the predictor’s response.

156 *4.1. Marginal-utility estimation*

Every candidate is represented by a cached feature vector h_j computed from its context, and the query and the current selected set are represented by h_Q and h_A . The selected-set encoder is permutation invariant, so the representation depends on the set and not on the order in which its elements were added. A cached model maps these representations to a utility score

$$\hat{m}_C(j \mid A) = g_C(h_Q, h_A, h_j), \quad (7)$$

157 which is inexpensive because it never evaluates the predictor. It is also
 158 incomplete: cached features describe the content of a candidate, not how the
 159 predictor reacts to it.

160 *4.2. Response-aware routing*

The cached score forms a shortlist

$$\mathcal{Q}_A = \underset{j \notin A}{\text{TopQ}} \hat{m}_C(j \mid A), \quad |\mathcal{Q}_A| = q \ll K, \quad (8)$$

which bounds the number of predictor evaluations per decision to q out of K candidates. For each retained candidate the predictor is evaluated with and without that candidate, and a compact response summary is formed,

$$r_{A,j} = \psi(f_A(x), f_{A \cup \{j\}}(x)). \quad (9)$$

The summary contains the level of the current prediction, the level after adding the candidate, and the mean, standard deviation, mean absolute value, normalized Euclidean norm, and maximum absolute value of the change. These statistics describe how the candidate changes the prediction, which is the quantity that enters the marginal utility through Eq. (5). The cached

representations are the anchor history, the candidate history, and the selected candidate histories with a validity mask. The response-aware model combines them with the cached representations,

$$\hat{m}_R(j \mid A) = g_R(h_Q, h_A, h_j, r_{A,j}), \quad j \in \mathcal{Q}_A. \quad (10)$$

161 4.3. Training and sequential inference

162 The model is trained on states sampled from the training episodes. A
 163 state is a uniformly random subset of the candidate pool whose size is drawn
 164 uniformly from 0 to $B - 1$, so training covers empty, partial and nearly
 165 complete selected sets; deployment states follow the router’s own greedy
 166 trajectory instead, and this difference between the training and deployment
 167 state distributions is the main approximation in the design. At each state the
 168 counterfactual loss difference of Eq. (4) provides the observed marginal for
 169 every candidate, so the labels are state-conditioned rather than standalone,
 170 and one state supplies $K - |A|$ regression targets together with all candidate
 171 pairs for the ranking term.

The encoder is a permutation-invariant set encoder of depth two with hidden and set width 64, and the utility head is a two-layer network of width 64. Training uses AdamW with learning rate 10^{-3} and weight decay 10^{-4} for 20 epochs with batches of 128 states; each target-run contributes 16,000 training states, four per episode, with 16 counterfactual labels each. The objective combines regression with a pairwise ranking term,

$$\mathcal{L} = \mathcal{L}_{\text{Huber}} + \beta \sum_{i,j: \tilde{m}_i > \tilde{m}_j} |\tilde{m}_i - \tilde{m}_j| \log(1 + \exp[-(\hat{m}_i - \hat{m}_j)]), \quad (11)$$

172 where the sum is taken over pairs within the same selection state and the
 173 weight is the utility gap of the pair. The gap weight assigns greater impor-

174 tance to ranking errors with larger utility consequences, while giving little
 175 weight to nearly tied candidates, and the regression term keeps the utility s-
 176 cale calibrated on states without a dominant alternative. The response-aware
 177 scorer is trained over candidate states sampled from the full training distribu-
 178 tion, where every candidate of a state carries a label and enters the pairwise
 179 term. At inference the scorer is evaluated only on the shortlist produced by
 180 the cached scorer.

181 At deployment the router starts from the empty set and repeats the fol-
 182 lowing steps: score all remaining candidates with the cached model, for-
 183 m the shortlist, evaluate the predictor on the shortlist, rerank it with the
 184 response-aware model, and add the highest-scoring candidate when its esti-
 185 mated marginal utility is positive. The procedure stops when no shortlist-
 186 ed candidate has a positive estimated utility or when the context budget
 187 is reached, and it is greedy: each decision maximizes the estimated one-
 188 step marginal utility within the shortlist. Section 5 analyzes the regret that
 189 screening and reranking introduce into this two-stage decision. Algorithm 1
 190 states the procedure.

191 **5. Decision Analysis**

192 R-MUR makes two decisions in sequence: which candidates enter the
 193 shortlist, and which shortlisted candidate is selected. The analysis separates
 194 the estimation error of the two stages, because the algorithm never evaluates
 195 the response model outside the shortlist.

Algorithm 1 Response-Aware Marginal-Utility Routing

Require: anchor S_0 , query x , candidate pool $\mathcal{C}$, budget B , shortlist size q ,

fixed predictor f

```
1:  $A \leftarrow \emptyset$ 
2: while  $|A| < B$  do
3:    $\hat{m}_C(j \mid A) \leftarrow g_C(h_Q, h_A, h_j)$  for all  $j \notin A$   $\triangleright$  cached screening
4:    $\mathcal{Q}_A \leftarrow \text{TopQ}_{j \notin A} \hat{m}_C(j \mid A)$ 
5:   for all  $j \in \mathcal{Q}_A$  do
6:      $r_{A,j} \leftarrow \psi(f_A(x), f_{A \cup \{j\}}(x))$   $\triangleright$  predictor response
7:      $\hat{m}_R(j \mid A) \leftarrow g_R(h_Q, h_A, h_j, r_{A,j})$ 
8:   end for
9:    $j^* \leftarrow \arg \max_{j \in \mathcal{Q}_A} \hat{m}_R(j \mid A)$ 
10:  if  $\hat{m}_R(j^* \mid A) \leq 0$  then
11:    break  $\triangleright$  stop: no shortlisted candidate is estimated useful
12:  end if
13:   $A \leftarrow A \cup \{j^*\}$ 
14: end while
15: return  $A$ 
```

196 *5.1. Shortlist-aware regret*

Let A be a fixed state, let $j^* = \arg \max_{j \notin A} m(j \mid A)$ be the best remaining candidate, let $j_Q^* = \arg \max_{j \in \mathcal{Q}_A} m(j \mid A)$ be the best candidate inside the shortlist, and let $\hat{j}$ be the candidate the router selects from the shortlist. The routing regret decomposes exactly into a screening term and a reranking

term,

$$\underbrace{m(j^\star | A) - m(\hat{j} | A)}_{\text{routing regret}} = \underbrace{m(j^\star | A) - m(j_Q^\star | A)}_{\text{screening regret}} + \underbrace{m(j_Q^\star | A) - m(\hat{j} | A)}_{\text{reranking regret}}. \quad (12)$$

Each term is controlled by the error of the corresponding stage.

Proposition 1 (Shortlist-aware one-step regret). *Suppose the cached score satisfies $|\hat{m}_C(j | A) - m(j | A)| \leq \epsilon_C$ for every remaining candidate, and the response-aware score satisfies $|\hat{m}_R(j | A) - m(j | A)| \leq \epsilon_R$ for every shortlisted candidate $j \in \mathcal{Q}_A$. Then the routing regret of Eq. (12) is at most*

$$m(j^\star | A) - m(\hat{j} | A) \leq 2\epsilon_C + 2\epsilon_R. \quad (13)$$

Proof. For the screening term, if $j^\star \in \mathcal{Q}_A$ the term is zero. Otherwise every shortlisted candidate has a cached score at least that of $j^\star$, because the shortlist consists of the q largest cached scores. Hence $\hat{m}_C(j_Q^\star | A) \geq \hat{m}_C(j^\star | A)$ and therefore $m(j_Q^\star | A) \geq m(j^\star | A) - 2\epsilon_C$. For the reranking term, $j_Q^\star$ is feasible for the estimated argmax, so $\hat{m}_R(j_Q^\star | A) \leq \hat{m}_R(\hat{j} | A)$, and two applications of the response bound give $m(j_Q^\star | A) - m(\hat{j} | A) \leq 2\epsilon_R$. $\square$

The bound in Eq. (13) is the statement that matches Algorithm 1: the first term is the price of screening with cached representations, the second is the price of estimating the response-aware score on the retained set. The decomposition also exposes the cost–accuracy trade-off of the shortlist: increasing q reduces the empirical screening regret and makes the best candidate more likely to reach the response-aware stage, while requiring more predictor evaluations. Section 6 measures both sides of that trade-off.

211 5.2. Stopping

212 At deployment, R-MUR stops when no shortlisted candidate has a posi-
213 tive estimated marginal utility. This stopping decision concerns the retained
214 set: candidates excluded by the cached screen are not evaluated by the
215 response-aware model. The shortlist-aware regret in Proposition 1 quantifies
216 the corresponding screening error, and the recall measurement in Section 6
217 reports how often the screen removes the best candidate.

218 5.3. Scope of the set-level statement

219 The one-step results above require no assumption on the set function.
220 A guarantee for a sequence of B selections would additionally require the
221 induced set utility to be monotone and submodular, so that approximate
222 greedy selection inherits the standard $(1 - 1/e)$ factor up to the accumulat-
223 ed estimation error. Context utility is not monotone in general, because a
224 context can increase the loss, and we do not assume submodularity in the ex-
225 periments. We therefore state the sequential guarantee as an open condition
226 rather than claiming it for R-MUR.

227 6. Experiments

228 6.1. Experimental setup

229 The controlled environment isolates the two mechanisms that make state
230 conditioning valuable. It fixes a context budget of four and uses 16 candi-
231 dates in the redundancy study; the candidate-scaling study varies the pool
232 size over $K = 4, 8, 16, 32, 64$ at a duplicate ratio of .50. Candidate value is
233 generated from a frozen subset-capable predictor, and Standalone Utility, the

234 cached-only router, and R-MUR use matched encoder and head capacity. All
 235 controlled results are averaged over five independent runs. The standalone
 236 oracle ranks candidates by the true standalone marginal and keeps that or-
 237 der fixed; the sequential oracle recomputes the true marginal after every
 238 selection. The gap between these two policies defines structural headroom.

239 Traffic forecasting provides a useful multi-context testbed because the
 240 prediction at one sensor can draw on correlated neighbouring sensors, and
 241 recent KBS models exploit increasingly rich spatiotemporal and cross-city
 242 dependencies [66–69]. Our experiments do not introduce a forecasting back-
 243 bone; the datasets are used to evaluate context-selection policies under a
 244 common fixed predictor.

245 The real-data study uses six standard traffic benchmarks: METR-LA,
 246 PEMS-BAY, PEMS03, PEMS04, PEMS07, and PEMS08. For each target
 247 sensor, a ridge predictor is fitted on that target’s training windows under a
 248 full, an empty, and a randomly masked subset of its own candidate pool, and
 249 is then held fixed for candidate evaluation, oracle computation, and router
 250 training; the predictor is therefore specific to the target and is refitted for
 251 every target the protocol evaluates. Candidate pools are constructed from the
 252 training segment with history length 12, forecast horizon 12, 16 candidates,
 253 and a context budget of four. The evaluation protocol fixes 32 evenly spaced
 254 target sensors per dataset and repeats each target with three independent
 255 runs, giving 576 target-run evaluations for the learned policies and 960 for
 256 the oracle policies. All methods are evaluated on the same pools, and the
 257 learned methods share encoder and head capacity. R-MUR uses a shortlist
 258 of four candidates, so a decision evaluates the predictor on one quarter of the

259 pool.

260 We report the prediction gain, that is, the loss reduction relative to
261 the anchor-only predictor; structural headroom H_{state} ; the negative-transfer
262 rate, that is, the fraction of evaluation episodes whose routed loss exceeds
263 the anchor-only loss; oracle recovery, that is, the ratio of learned gain to
264 sequential-oracle gain; and the number of predictor evaluations per decision.
265 Standalone Utility ranks candidates by a predicted standalone marginal,
266 and the cached-only router is the same state-conditioned model without
267 predictor-response features. Four further rules are evaluated on the same
268 episodes, candidate pools and frozen predictor, all of them content-based
269 and none of them trained or evaluating the predictor during selection: facili-
270 ty location, which greedily maximises submodular coverage of the candidate
271 pool; the same rule with an added relevance term; greedy determinantal
272 point process selection, which maximises the log-determinant of a similarity
273 kernel; and k -center selection, which maximises the minimum distance to
274 the selected set. Differences across methods are reported with hierarchical
275 bootstrap intervals over 5000 replicates that resample targets and then runs
276 within a target, so both sources of variation enter the interval.

277 6.2. When does state conditioning matter?

278 In the controlled redundancy study, structural headroom grows mono-
279 tonically with the duplicate ratio. At zero redundancy the standalone and
280 sequential oracles coincide, so a perfect standalone ranking loses nothing. As
281 redundancy increases, the gap reaches .286, and at duplicate ratios .50 and
282 .75 the learned router exceeds the standalone oracle even though that oracle
283 uses the true standalone utilities. Conditioning on the selected set therefore

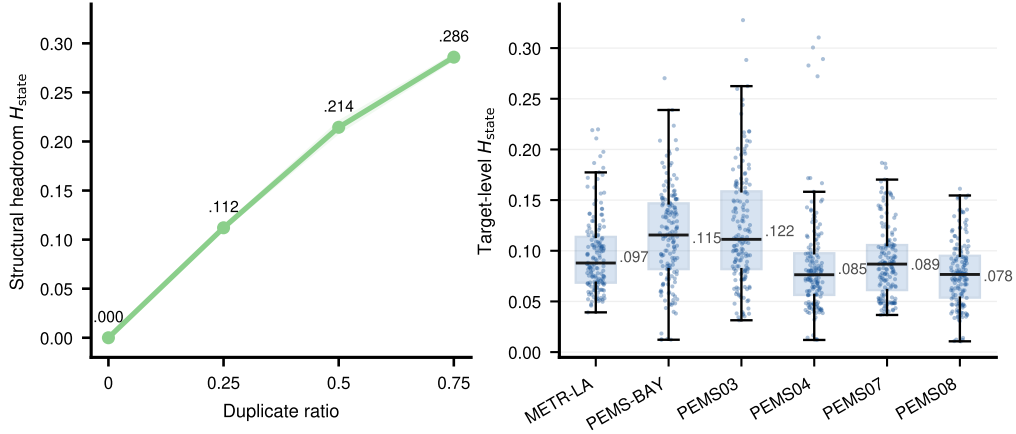


Figure 2: Structural headroom. Left: the gap between a perfect standalone ranking and a sequential oracle grows with the duplicate ratio, and the learned router exceeds the standalone oracle once redundancy is high. Right: target-level headroom on the six benchmarks, one point per target.

284 creates decision value beyond what standalone scoring provides. The per-run
 285 ladder, including the gains of both oracles, is reported in the supplementary
 286 material.

287 The same effect appears on real benchmarks. Over six datasets, 32 fixed
 288 targets per dataset, and five independent runs, every one of the 960 target-run
 289 evaluations has positive structural headroom. Figure 2 shows that the target-
 290 level distributions are concentrated rather than driven by a few sensors: the
 291 per-dataset means range from .078 to .122 and the medians from .076 to .116.
 292 Headroom of this size means that a router which only reproduces a perfect
 293 standalone ranking leaves a measurable part of the available loss reduction
 294 unused on every dataset and every target we evaluated. Table 1 reports the
 295 ratio together with the unnormalised gap, which ranges from .005 to .019 in
 296 loss units.

Table 1: Structural headroom per dataset over 32 fixed targets and 5 independent runs (960 target-run evaluations). The ratio and the unnormalised difference are reported together; every evaluation has positive headroom.

Dataset	Mean H_{state}	Median H_{state}	Mean gap	Positive fraction
METR-LA	.0966	.0879	.0143	1.000
PEMS-BAY	.1153	.1155	.0188	1.000
PEMS03	.1221	.1113	.0071	1.000
PEMS04	.0848	.0763	.0052	1.000
PEMS07	.0891	.0868	.0057	1.000
PEMS08	.0778	.0766	.0050	1.000

297 6.3. Multi-target routing performance

298 Table 2 compares sixteen deployable selection policies under the multi-
 299 target protocol. It contains the learned policies, which are trained separately
 300 for each target-run on that target’s training episodes; four content-based
 301 rules that train nothing, namely facility location, relevance-weighted facil-
 302 ity location, greedy DPP, and k -center selection; and five acquisition rules
 303 adapted from feature selection and active feature acquisition, which choose
 304 one set per target on the validation split and then apply it to every test
 305 episode. The fixed-set rules estimate how much of the gain is available with-
 306 out conditioning a decision on the state of an individual episode. Within
 307 a run all methods share the same candidate pool, the same fixed predictor
 308 and the same evaluation episodes; the adaptation of each fixed-set rule is
 309 described in the supplementary material. R-MUR obtains the best average
 310 rank (1.83) and the lowest negative-transfer rate (.335) of the sixteen meth-

ods compared, and it is the best response-free method on every dataset. It
 is not the best method on every dataset, and we state where it is not: the
 trivial policy of using all sixteen candidates exceeds it on PEMS04 (+.0365
 against +.0357) and PEMS08 (+.0359 against +.0333), and the one-shot
 response-aware variant exceeds it on PEMS04, PEMS07 and PEMS08.
 Paired against pooling, under target-level paired intervals R-MUR is signifi-
 cantly ahead on METR-LA (+.0067 [+0.0041, +.0094]), PEMS-BAY (+.0303
 [+0.0203, +.0427]) and PEMS07 (+.0041 [+0.0015, +.0070]), is unresolved on
 PEMS03 (+.0013 [−.0004, +.0030]) and PEMS04 (−.0008 [−.0050, +.0025]),
 and is significantly below on PEMS08 (−.0026 [−.0042, −.0010]). The
 strongest competitor on every benchmark is the trivial policy of using every
 candidate, not any of the specialised subset rules. The fixed-set rules are
 the closest competitors on gain, and their harm rate is higher than that of
 R-MUR even though they commit to a single set per target: tuning a set
 on validation protects its average gain, not the individual episode, which is
 again what per-episode conditioning provides. The content-based rules that
 select by coverage or diversity alone, like relevance and maximum marginal
 relevance, are negative on most datasets: covering the candidate pool more
 evenly does not make the selected contexts more useful to the predictor.

The comparison forms a ladder in which each rung answers a different
 question. Relevance, MMR, facility location, DPP and k -center all decide
 from cached content alone, and their failure shows that neither similarity, nor
 diversity, nor coverage is a reliable proxy for downstream predictive value.
 The fixed-set rules are stronger than the content-based ones because they
 are tuned against the predictor, and greedy validation selection is the closest

competitor in the table; it still remains below R-MUR on every dataset, so choosing the best set offline is not a substitute for deciding per episode. That gap, small and consistent, is the value that state conditioning adds. Standalone Utility, which learns the value of a candidate in isolation, recovers part of the gap but stays below the cached-only router on average, so a fixed candidate value is not sufficient. The cached-only router conditions on the selected set but never evaluates the predictor, and R-MUR improves on it on every dataset, which isolates the contribution of the predictor response. The resulting order,

content-based rules < Standalone Utility $\lesssim$ cached-only router < fixed-set acquisition < R-MUR

330 separates the four design decisions instead of reporting a single ranking:
 331 choosing informative content, estimating a candidate’s value, conditioning
 332 that value on the selected set, and finally observing the predictor response.

333 Table 3 reports the two contrasts that separate the components of
 334 the method. The improvement over Standalone Utility shows that state-
 335 conditioned marginal utility contains decision information that a fixed can-
 336 didate score does not; the improvement over the cached-only router shows
 337 that evaluating the predictor on the shortlist adds information beyond a
 338 state-conditioned score built from cached context features. Both differences
 339 are positive on every dataset, and the hierarchical bootstrap intervals exclude
 340 zero everywhere. The mean gain is positive on 81% to 100% of targets per
 341 dataset, and 97% to 100% of targets exceed the standalone baseline, so the
 342 improvement extends across prediction targets rather than being driven by a
 343 single sensor. Oracle recovery ranges from .10 to .53. METR-LA shows the
 344 widest target-level variation and the lowest recovery, while its dataset-level

Table 2: Deployable selection methods under the multi-target protocol. Values are prediction gains averaged over 32 fixed targets per dataset and over 3 independent runs. The first group holds the reference and learned policies, the second content-based subset rules, and the third fixed-set acquisition rules. The average rank is computed over all listed methods, and the final column is the mean negative-transfer rate.

Method	METR-LA	PEMS-BAY	PEMS03	PEMS04	PEMS07	PEMS08	Avg. rank	Neg. transfer
Relevance	-.0165	-.0199	-.0191	.0071	-.0155	.0074	15.17	.5856
MMR	-.0180	-.0249	-.0102	.0098	-.0085	.0075	14.67	.5703
Standalone Utility	.0024	.0272	.0149	.0248	.0152	.0246	7.00	.4367
Cached-only router	.0042	.0284	.0154	.0250	.0150	.0246	6.33	.3944
R-MUR	.0146	.0557	.0269	.0357	.0284	.0333	1.83	.3346
Pool all 16	.0080	.0254	.0256	.0365	.0242	.0359	3.50	.4270
Response-aware one-shot	-.0017	.0502	.0263	.0382	.0302	.0353	2.67	.4065
Facility location	-.0210	.0077	.0088	.0139	.0074	.0171	10.17	.4733
DPP greedy	-.0151	.0053	.0070	.0131	.0058	.0120	10.33	.4912
Facility location + relevance	-.0171	-.0071	-.0017	.0123	-.0023	.0129	11.67	.5321
k-center	-.0181	-.0140	.0044	.0123	.0031	.0095	12.67	.5168
Greedy validation set	.0088	.0403	.0197	.0308	.0182	.0297	3.83	.3850
Adaptive- k relevance	-.0040	-.0076	-.0061	.0087	-.0052	.0083	12.83	.5559
Cluster-based selection	-.0162	-.0127	-.0055	.0087	-.0002	.0125	12.50	.5286
Two-step lookahead set	.0089	.0402	.0197	.0308	.0182	.0297	4.17	.3884
Gradient influence	.0004	.0349	.0131	.0284	.0144	.0277	6.67	.4157

gain remains positive.

Across the six benchmarks, R-MUR achieves the best average rank among all sixteen evaluated deployable policies and is significantly ahead of the strongest competitor on three of the six, unresolved on two and losing on one, so these results establish it as the strongest *subset-selecting* method in our unified evaluation of relevance-, diversity-, coverage-, standalone-utility-, and state-conditioned selection. The two oracle policies serve as structural references and are excluded from the deployable ranking: the standalone oracle shows what a fixed ranking can reach, and the sequential oracle what sequential information can reach.

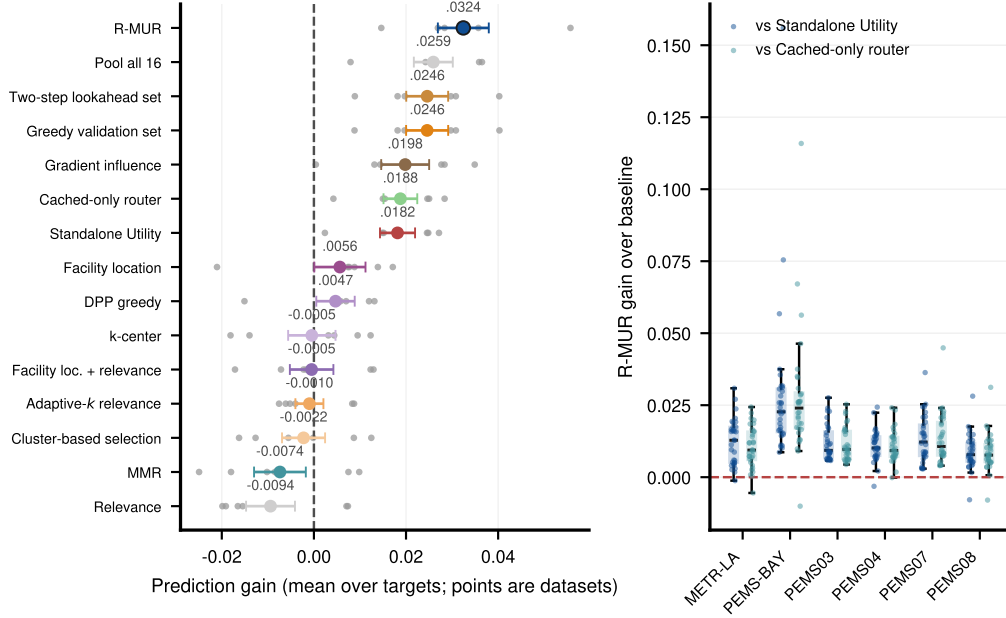


Figure 3: Multi-target comparison over fifteen deployable selection policies, including the trivial pool-everything policy that is the strongest competitor (the one-shot response ablation appears in Table 2 only). Left: mean prediction gain across the six benchmarks, with the policies ordered by that mean; small points are the per-dataset target means and the interval is the standard error over datasets. Right: distribution over the 32 fixed targets of the R-MUR gain over Standalone Utility and over the cached-only router.

355 6.4. Why does R-MUR improve selection?

356 Two mechanisms explain the result. The first is that larger candidate
 357 pools compress the decision margin. In the controlled scaling study, the
 358 absolute utility error improves as K increases, but the standard deviation
 359 of the true marginals falls faster: normalized error rises from .403 to .739,
 360 the positive top-two margin falls from .220 to .024, and the median ratio of
 361 maximum candidate error to that margin rises from 1.17 to 3.40. Selection
 362 accuracy falls from .884 to .147, and one-step regret rises from .018 to .067 at

Table 3: Multi-target consistency of R-MUR over 32 fixed targets per dataset and 3 independent runs. Differences are computed per target and then averaged; brackets are 95% hierarchical bootstrap intervals that resample targets and then runs within a target. Positive targets is the fraction of targets whose mean gain is positive. Oracle recovery is the pooled ratio of R-MUR gain to sequential-oracle gain.

Dataset	Δ vs Static	Δ vs Cached	Positive targets	> Static	> Cached	Oracle recovery
METR-LA	.0123 [.0082, .0168]	.0104 [.0069, .0142]	0.812	0.969	0.969	.099
PEMS-BAY	.0285 [.0202, .0401]	.0272 [.0199, .0377]	0.938	1.000	0.969	.309
PEMS03	.0121 [.0099, .0145]	.0115 [.0095, .0136]	1.000	1.000	1.000	.447
PEMS04	.0109 [.0089, .0129]	.0107 [.0088, .0127]	1.000	0.969	0.969	.534
PEMS07	.0132 [.0104, .0161]	.0134 [.0105, .0167]	1.000	1.000	1.000	.456
PEMS08	.0088 [.0067, .0109]	.0087 [.0065, .0110]	1.000	0.969	0.969	.516

363 $K = 16$ and remains elevated at .049 at $K = 64$. Pointwise regression error
364 therefore does not describe decision quality, which is why R-MUR is trained
365 with a gap-weighted ranking term in addition to regression.

366 The second mechanism is the information carried by the predictor re-
367 sponse. With cached context features only, the response-feature comparison
368 reaches a Spearman correlation of .164 and a positive-utility area under the
369 receiver operating characteristic curve of .649. Adding compact response
370 summaries raises them to .390 and .834, improves top-1 accuracy from .148
371 to .259, and reduces one-step regret from .0736 to .0569. Full forecast tra-
372 jectories add little beyond the compact summaries. Figure 4 collects the two
373 mechanisms.

374 Table 4 reports the component ablation under the multi-target protocol.
375 Using the whole pool without selection already improves over the anchor,
376 and it exceeds the cached-only router, which is state-conditioned but never
377 evaluates the predictor. R-MUR improves on both while evaluating the pre-

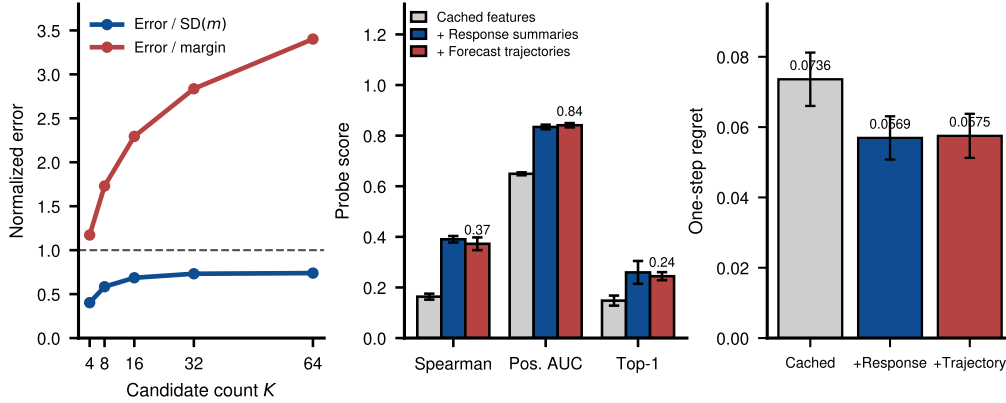


Figure 4: Mechanisms. Left: as the candidate pool grows, the decision margin shrinks faster than the pointwise error, so selection accuracy falls. Middle: rank correlation, positive-utility AUC, and top-1 accuracy for cached features, compact response summaries, and full forecast trajectories. Right: one-step regret of the same three input sets.

Table 4: Component ablation under the multi-target protocol, averaged over all targets and runs of the primary multi-target evaluation, which comprises three independent runs per target. Predictor evaluations is the share of the candidate pool that receives a predictor evaluation per decision.

Variant	Mean gain	Neg. transfer	Contexts used	Predictor evaluations
Anchor only	.0000	.0000	.00	0
Pool-all (no selection)	.0259	.4270	16.00	100%
Cached-only router	.0188	.3944	3.18	0
R-MUR	.0324	.3346	3.05	25%

dictor on a quarter of the pool and using about three contexts per episode.
The gap between the cached-only router and R-MUR is the contribution
of the predictor-response stage, and it is larger than the gap between the
cached-only router and a fixed standalone score.

Table 5: One-step routing regret decomposed into the screening and reranking terms of Eq. (12), in true marginal-utility units, at the deployed shortlist size $q = 4$. The final column is the screening share.

Dataset	Screening	Reranking	Total	Screening share
METR-LA	.0319	.0326	.0645	.49
PEMS-BAY	.0298	.0358	.0656	.45
PEMS03	.0083	.0073	.0156	.53
PEMS04	.0080	.0070	.0151	.53
PEMS07	.0092	.0077	.0169	.55
PEMS08	.0077	.0071	.0149	.52

382 6.5. Routing budget and residual error

383 Oracle recovery is between .10 and .53 (Table 3), so a substantial part
384 of the sequential-oracle gain is not realised. The regret decomposition of
385 Eq. (12) locates it. At the deployed shortlist size, the screening term accounts
386 for 45% to 55% of the one-step regret and the reranking term for the rest
387 (Table 5). The loss is therefore not dominated by one stage: the cached screen
388 misses the truly best candidate often enough to matter, and the response-
389 aware model misranks the candidates it does see almost as often.

390 Table 6 shows how the screening term responds to the shortlist size. Re-
391 call of the truly best candidate is .09–.12 at $q = 1$, .29–.33 at the deployed
392 $q = 4$, and .52–.57 at $q = 8$, and the mean screening regret falls from .020–
393 .079 to .003–.017 over the same range. Screening is thus the stage where
394 additional predictor evaluations buy the most, which is the quantitative ver-

Table 6: Cached screening as a function of the shortlist size. Each cell reports the recall of the truly best candidate and the mean screening regret in true marginal-utility units.

Dataset	$q = 1$	$q = 2$	$q = 4$	$q = 8$	$q = 16$
METR-LA	0.09 / .070	0.17 / .048	0.29 / .032	0.52 / .017	1.00 / .000
PEMS-BAY	0.10 / .079	0.18 / .047	0.31 / .030	0.55 / .017	1.00 / .000
PEMS03	0.10 / .021	0.17 / .013	0.30 / .008	0.55 / .004	1.00 / .000
PEMS04	0.10 / .021	0.18 / .013	0.31 / .008	0.57 / .003	1.00 / .000
PEMS07	0.10 / .024	0.17 / .015	0.30 / .009	0.54 / .004	1.00 / .000
PEMS08	0.12 / .020	0.20 / .013	0.33 / .008	0.57 / .004	1.00 / .000

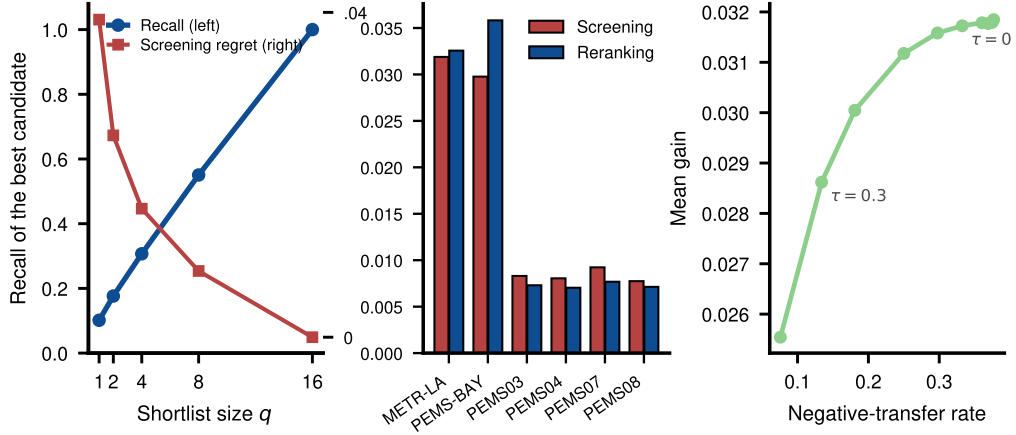


Figure 5: Shortlist diagnostics. Left: recall of the truly best candidate and screening regret as the shortlist grows. Middle: the one-step routing regret decomposed into its screening and reranking terms at the deployed shortlist size. Right: the gain–harm trade-off obtained by raising the stopping threshold.

395 sion of the trade-off behind the shortlist: each doubling of q roughly doubles
396 the probability that the best candidate is available for reranking, at a pro-
397 portional cost in predictor evaluations.

398 The shortlist bounds the cost of a decision, and the stopping threshold
 399 decides how much of it is used. At $q = 4$ the predictor is evaluated on one
 400 quarter of the pool, and Table 6 shows what that budget buys. The stopping
 401 threshold is the second part of the cost, and it also controls how often routing
 402 is harmful. Table 7 sweeps the threshold τ on the estimated marginal utility.
 403 At the deployed threshold of zero the router keeps a mean gain of .0317 with
 404 a negative-transfer rate of .332; raising the threshold to .3 keeps .0286 of
 405 the gain, reduces negative transfer to .134 and uses about one context per
 406 episode; at .5 the router keeps .0255 of the gain with a negative-transfer rate
 407 of .076.

408 The same table explains the residual negative transfer. At the deployed
 409 threshold the positive-utility decision has precision .61 and recall .79: the
 410 response-aware score identifies most useful candidates but over-predicts use-
 411 fulness, so a sign threshold is not a calibrated stopping rule. The sweep expos-
 412 es a clear operating trade-off between predictive gain and negative transfer,
 413 and it turns the stopping rule into an explicit operating point rather than
 414 a fixed constant. We report $\tau = 0$ in all other results so that no method is
 415 tuned on the evaluation targets.

416 7. Discussion

417 Two quantities organise the results. Structural headroom measures the
 418 opportunity that state conditioning creates: it is the loss reduction available
 419 to a sequential rule over a perfect standalone ranking, and it is a property of
 420 the prediction task rather than of a particular router. The controlled study
 421 shows that redundancy creates this opportunity, and the multi-target study

Table 7: Stopping threshold sweep, averaged over the six datasets. Raising the threshold on the estimated marginal utility trades gain for a lower negative-transfer rate and fewer contexts. Coverage is the fraction of episodes on which the router selects at least one context. At the deployed threshold of zero the positive-utility decision has precision 0.61 and recall 0.79, where precision is the fraction of selected candidates with positive true marginal utility and recall is the fraction of positive-utility candidates that are selected.

Threshold τ	Mean gain	Neg. transfer	Contexts	Coverage
-0.40	.0318	.377	3.42	1.00
-0.20	.0318	.375	3.40	1.00
-0.10	.0318	.369	3.35	0.98
-0.05	.0318	.360	3.28	0.97
0.00	.0317	.332	3.05	0.91
0.05	.0316	.298	2.44	0.84
0.10	.0312	.250	1.93	0.74
0.20	.0300	.181	1.34	0.58
0.30	.0286	.134	0.98	0.47
0.50	.0255	.076	0.56	0.32

422 shows that it is positive on every dataset and every target we evaluated.
423 Predictor response and ranking quality then determine how much of that
424 opportunity a learned router can realise. Cached representations of context
425 content do not carry the information that the marginal utility depends on,
426 while a compact summary of the predictor’s response does; and as candidate
427 pools grow, the utility gap between the best alternatives shrinks, so a model

428 with stable average error becomes less able to identify the best candidate.
429 Both effects are visible in the results and both are addressed by the design
430 of R-MUR.

431 The multi-target results also show that realisation is uneven. METR-LA
432 converts the least headroom into realised gain, with the lowest oracle recovery
433 and the fewest targets whose gain is positive, while PEMS-BAY shows the
434 widest spread across targets. Structural opportunity alone therefore does not
435 determine how much value a router obtains, which is why the two quantities
436 are reported separately.

437 R-MUR uses greedy one-step selection, so it does not capture comple-
438 mentarity that emerges only from selecting a group of individually unhelpful
439 contexts.

440 *7.1. Deciding whether to route at all*

441 Because the improvement is not uniform across benchmarks, a practi-
442 tioner needs a way to decide before committing. The measurement protocol
443 of the companion studies supplies one, and its operating characteristics are
444 measured rather than asserted: on pilots of n targets drawn from six deploy-
445 ments, a pilot of 24 targets reproduces the full-deployment verdict 96% of
446 the time, with 98% sensitivity and 96% specificity for the decision to route
447 at all, while sixteen targets give 84% and 86%. The direction of the effect is
448 much easier to recover than its significance. The procedure is to label a pilot
449 of roughly two dozen targets, run the protocol unchanged on it — screen-
450 ing, budget, ranking objective, and both the router and the pool-everything
451 baseline — and route only if the paired target-level interval excludes zero
452 above. Cheaper substitutes do not work, and each is reported as a nega-

453 tive in the companion work: a fitted predictability probe is indistinguishable
454 from chance, structural headroom does not order the benefit, and the cached
455 screen is uncorrelated with the true marginal exactly when the decision is
456 close.

457 A pilot is also labelled earlier than the period it is used for, and that shift
458 is measured rather than assumed: scoring the same 32 targets on the calibra-
459 tion and test periods of six deployments gives a within-dataset correlation of
460 $+0.596$ between the two, per-target sign agreement of 81%, and dataset-level
461 verdict agreement of four of six — direction transfers, the significance call
462 less well, and no pilot recommended routing where routing turned out to
463 hurt.

464 The pilot can also be audited against itself, and the audit is worth more
465 than the raw verdict: splitting it in half, pilots whose halves both resolve in
466 the same direction agree with the full-sample verdict 99.0% of the time at
467 sixteen targets against 78.1% for the rest. The check certifies rather than
468 produces a verdict — it holds for only about a third of pilots — so the rule
469 is to act when it passes and to enlarge the pilot when it does not.

470 Training also requires counterfactual predictor evaluations to construct
471 the marginal-utility targets. These costs arise offline, while deployment lim-
472 its predictor evaluations through cached screening. The current experiments
473 study subset-capable predictors on traffic forecasting; extending the same de-
474 cision formulation to other predictor families and context-selection domains
475 is a natural direction for further study.

476 **8. Conclusion**

477 We formulated context selection through the set-conditioned marginal re-
478 duction in end-task loss. This decision object exposes structural headroom
479 created by state conditioning, and it separates that opportunity from the ac-
480 curacy with which a router can estimate it. R-MUR combines cached screen-
481 ing with compact predictor-response summaries and a gap-weighted ranking
482 objective. Across six standard traffic benchmarks, R-MUR improves learned
483 context selection over standalone and cached baselines, and a multi-target
484 study shows that structural headroom persists across sensors. Predictor re-
485 sponse and decision margin are the key quantities for turning that structural
486 opportunity into learned routing gains.

487 **CRedit authorship contribution statement**

488 **Xinling Wen:** [roles]. **Jiahao Zhang:** [roles]. **Yu Chen:** [roles]. All
489 authors have read and approved the final manuscript.

490 **Declaration of competing interest**

491 The authors declare that they have no known competing financial inter-
492 ests or personal relationships that could have appeared to influence the work
493 reported in this paper.

494 **Data availability**

495 The traffic benchmarks used in this study are publicly available. The
496 candidate-pool construction, routing protocol, model configuration, and eval-

497 uation procedure required to reproduce the reported numbers are described
498 in the manuscript and supplementary material.

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